

Sukhraj Singh Nandhra

+91-9667706428 — sn699@snu.edu.in — Github — Portfolio

Education

Shiv Nadar Institute of Eminence, Greater Noida

GPA: 9.21

B.Tech in Computer Science Engineering

Dean's List: Monsoon 2023, 2025; Spring 2025, 2026

KIIT World School, New Delhi

Class XII (CBSE): 95.2%

Class X (CBSE): 98.8%

Internships

Hewlett Packard Enterprise — CPP3 Intern

Feb 2026 - July 2026

- Developed an always-on security platform for containerized HPC workloads using Python, Docker, ZeroMQ, and SQLite for continuous telemetry and threat detection.
- Integrated runtime security tools including Falco, Suricata and Docker event monitoring to detect container escapes, privilege escalation, network threats and configuration drift.
- Built a risk-based detection pipeline with event correlation, automated container remediation, forensic evidence capture aligned with NIST compliances.

Hewlett Packard Enterprise — CPP2 Intern

May 2025 - Jan 2026

- Integrated the Dolos plagiarism detection engine with the DMOJ online judge to automate similarity analysis of programming submissions.
- Developed Python automation for submission extraction, report processing, and similarity score generation, streamlining the plagiarism detection workflow.

Projects

Crux — Local Linux AI Assistant

Github

- Built an offline Linux CLI assistant using Retrieval-Augmented Generation (RAG) over Linux man pages with Ollama (Qwen2.5), ChromaDB, and Sentence Transformers.
- Implemented semantic retrieval and command generation from natural language, enabling grounded responses and Linux command synthesis without internet or external APIs.
- Designed a command safety pipeline that classifies generated commands before execution and requires explicit user approval for potentially destructive operations.

Palladium — Cross-Source News Intelligence Platform

Palladium

- Built a news intelligence platform that clusters articles from 15 international RSS feeds using sentence-transformer embeddings and agglomerative clustering, comparing publisher coverage with spaCy NER and TF-IDF.
- Developed a deterministic analysis engine with LexRank summarization, entity consensus, coverage scoring, and a modular Python/Flask backend for automated news analysis.

Positions of Responsibility

Shiv Nadar University — Teaching Assistant

Sept 2025 - May 2026

- Supported faculty in delivering lectures and lab sessions on Data Structures and DBMS.

Content Team Lead — ACM SNU

Sept 2025 - May 2026

- Led the content team in producing event reports and promotional material for workshops and programming events.
- Coordinated with technical and management teams to plan workshops and coding events.

Technical Skills

Languages: C, C++, Python, JavaScript, SQL

Frameworks/Libraries: FastAPI, Flask, Django, Typer, Rich, Sentence-Transformers

AI & Databases: Ollama, RAG, ChromaDB, SQLite, MySQL, Supabase

Tools: Linux, Git, Docker, Bash